Simple Harmonic Motion (SHM)

Harmonic motion

- A motion which repeats in equal intervals of time is termed as a periodic motion.

- If the periodic motion occurs back and forth over the same path, then it is termed oscillatory or harmonic.

- Examples include swinging pendulum of a grandfather clock, and the vibrations produced in an organ pipe.

Important terms in periodic motion

- **Amplitude** (A) – maximum displacement from the equilibrium position. The SI unit is metre.

- **Period** (T) – Time taken to complete one oscillation or cycle. The SI unit is second.

- **Frequency** (f) – Number of oscillations or cycles in a unit of time. The SI unit is hertz (Hz) or cycle/s.

$$f = \frac{1}{T}$$

- **Angular frequency** (ω) – Angular displacement per second. Its unit is rad/s.

$$\omega = 2\pi f$$

Simple harmonic motion (SHM)

- **Simple harmonic motion (SHM)** is a kind of oscillation where the force is directly proportional to the displacement of the mass from the equilibrium position, i.e.

$$F = -kx$$

where k is a constant.

- This force is also called as the **restoring force** as it restores an object to the equilibrium position.

- The negative sign indicates that the direction of the force and the displacement are opposite.

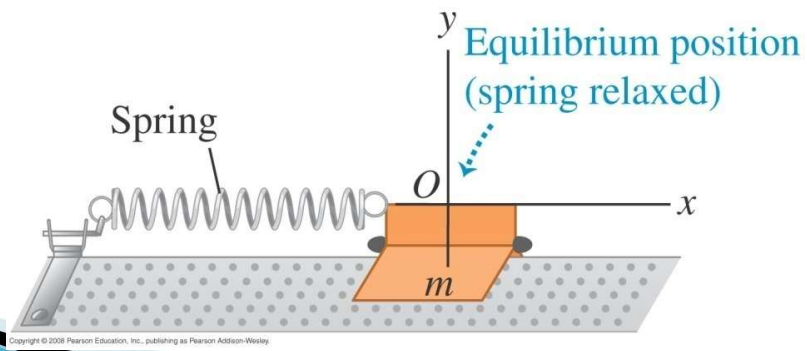
Application of simple harmonic motion (SHM)

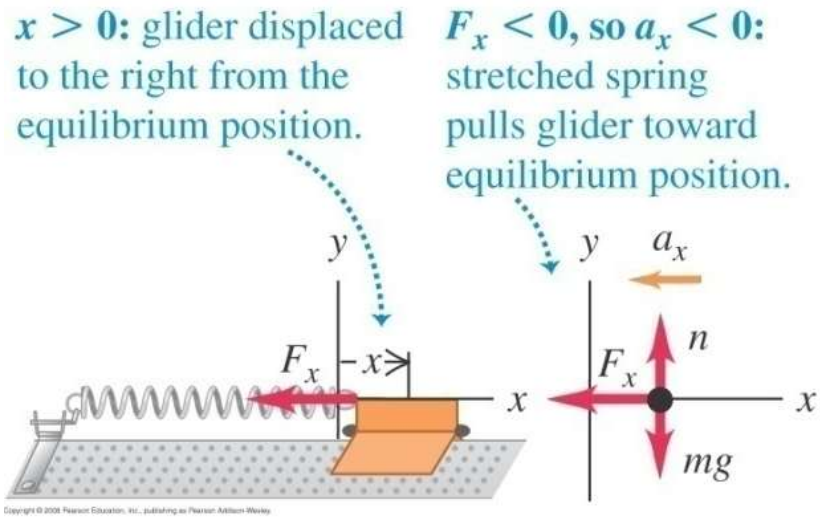
- In many systems the restoring force is approximately proportional to displacement if the displacement is small.
- Examples include a spring-mass system and alternating electric current flowing in a circuit.
- Therefore, SHM can be used as an approximate mode.
- Note that not every periodic motion is SHM.

Spring-mass system – an example of SHM

- A simple spring-mass system comprises an object of mass m, attached to a light spring, on a frictionless horizontal guide.

- If the spring is stretched or compressed from the equilibrium position *O* (taken as the origin), a net force is exerted ***by*** the spring ***on*** the object in the horizontal direction to bring the object back to *O*. animation





- We can write the second law of Newton for a spring-mass system without damping forces as

$$F = ma = -kx$$

$$a = \frac{d^2x}{dt^2} = \frac{-k}{m}x$$

- In the spring-mass system, k is force constant or spring constant and its unit is N m^{-1}.

- The acceleration is **not** constant as it depends on x.

- A body that undergoes SHM is known as a simple ***harmonic oscillator.***

Solution of the differential equation

- The solution of the differential equation

$$\frac{d^2x}{dt^2} = \frac{-k}{m}x$$

is obtained by trial-error and the solution is a sinusoidal function. $x = A\cos(\omega t + \phi)$

where A is the amplitude and ϕ is the phase angle.

- If we substitute the above in the differential equation we get

$$\frac{d^2A\cos(\omega t+\phi)}{dt^2} = \frac{-k}{m}A\cos(\omega t+\phi)$$

$$\omega^2 = \frac{k}{m}$$

$$\text{or } \omega = \sqrt{\frac{k}{m}}$$

- The ***phase*** angle ϕ tells us at what point in the cycle the motion was at $t = 0$.
- We can also use sine function instead of cosine for the displacement using the identity

$$\cos\alpha = \sin\left(\alpha + \frac{\pi}{2}\right)$$

as the displacement x is a periodic sinusoidal function of time.

- Since the values of sine or cosine functions lie between –1 to 1, the displacement varies between –A to A.

- ω is the angular frequency and is determined by the spring constant k and the mass of the object m.
- ω is related to frequency and time period as

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$T = \frac{1}{f} = 2\pi\sqrt{\frac{m}{k}}$$

- Note that the period or frequency is independent of the amplitude for SHM.

Velocity and acceleration in SHM

- The velocity for a harmonic oscillator can be obtained as

$$v_x = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

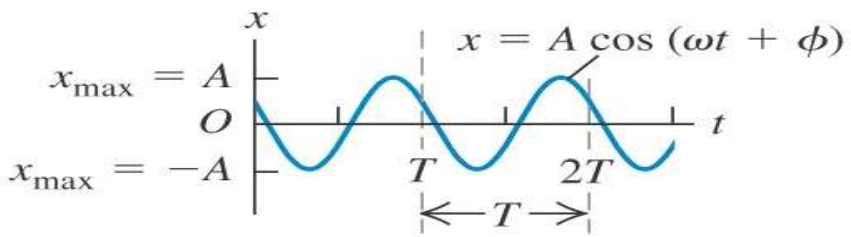
- The acceleration can be obtained as

$$a = \frac{dv_x}{dt} = \frac{d^2x}{dt^2} = -\omega^2 A \cos(\omega t + \phi)$$

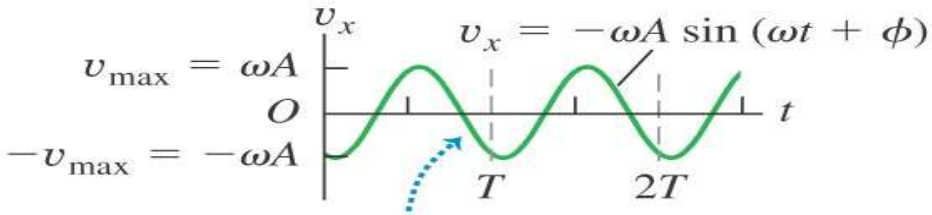
- The velocity oscillates between $-\omega A$ and $+\omega A$ whereas the acceleration takes values between $-\omega^2 A$ and $+\omega^2 A$.

Graphs of displacement, velocity and acceleration

(a) Displacement x as a function of time t

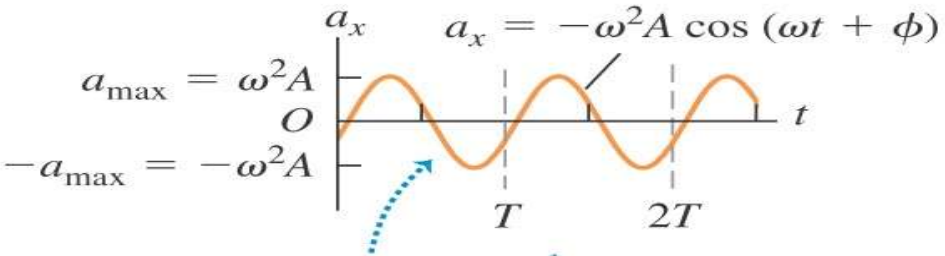


(b) Velocity v_x as a function of time t



The v_x-t graph is shifted by $\frac{1}{4}$ cycle from the x-t graph.

(c) Acceleration a_x as a function of time t



The a_x-t graph is shifted by $\frac{1}{4}$ cycle from the v_x-t graph and by $\frac{1}{2}$ cycle from the x-t graph.

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Determining amplitude and phase angle from initial conditions

- Let the initial position be x_0 and the initial velocity be v_{0x}.

- Putting t = 0 in the displacement and velocity equations gives

$$x_0 = A\cos(\phi)$$

$$v_{0x} = -\omega A\sin(\phi)$$

$$A = \sqrt{x_0^2 + \frac{v_{0x}^2}{\omega^2}}$$

$$\phi = \tan^{-1}\left(-\frac{v_{0x}}{\omega x_0}\right)$$

Example 1

An ultrasonic transducer (a kind of loudspeaker) used for medical diagnosis oscillates at a frequency of 6.7 MHz. How much time does each oscillation take and what is the angular frequency?

$$T = \frac{1}{f} = \frac{1}{6.7 \times 10^{6}\,\text{Hz}}$$

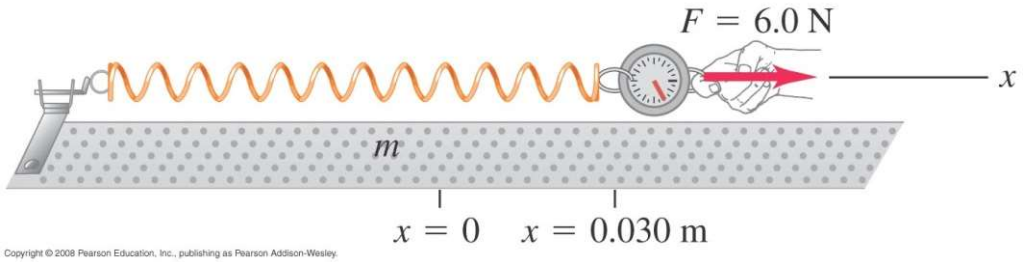
$$= 1.5 \times 10^{-7}\,\text{s}$$

$$\omega = 2\pi f = 4.2 \times 10^{7}\,\text{rad/s}$$

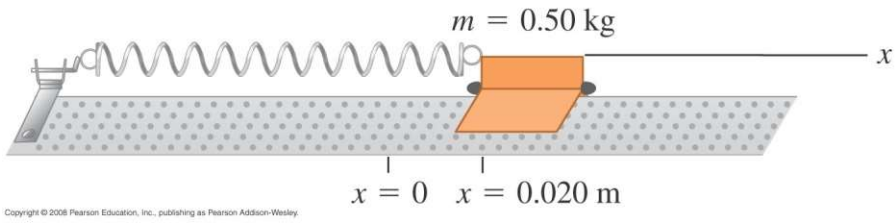
Example 2

A spring is mounted horizontally with its left end stationary. By attaching a spring balance to the free end and pulling toward the right we determine that the stretching force is proportional to the displacement and that a force of 6.0 N causes a displacement of 0.030 m.

Then the spring balance is replaced by a 0.50-kg glider which is pulled through a distance of 0.020 m along a frictionless air track and released. Find the force constant, angular frequency, frequency and period of oscillation.



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Example 3

Referring to example 2 with force constant = 200 N/m and mass 0.50 kg, if we give the body an initial displacement of +0.015 m and velocity of +0.40 m/s,

a) Find the period, amplitude and phase angle of the motion.

b) Write the equations for the displacement, velocity and acceleration as functions of time.

Energy in SHM of a spring-mass system

- The total mechanical energy is conserved as the force exerted by the spring is conservative.

- Neglecting the mass of the spring, the total mechanical energy consists of the kinetic energy of the object and the potential energy of the spring, i.e.

$$
\begin{aligned}
E &= K + U \\
&= \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \\
&= \frac{1}{2}kA^2 \text{ (where A is the amplitude)} \\
&= \text{constant}
\end{aligned}
$$

Energy in SHM of a spring-mass system

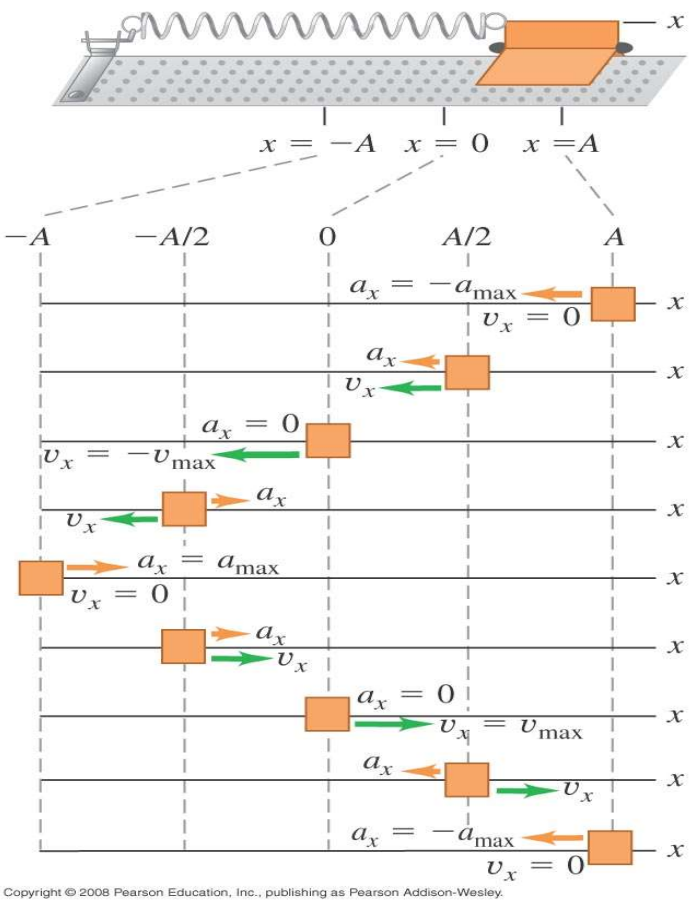
- The energy equation can be used to calculate the object velocity at a given displacement, viz:

$$v_x = \pm\sqrt{\frac{k}{m}}\sqrt{A^2 - x^2}$$

- The maximum velocity occurs at x = 0, i.e

$$v_{max} = \pm\sqrt{\frac{k}{m}}A = \pm\omega A$$

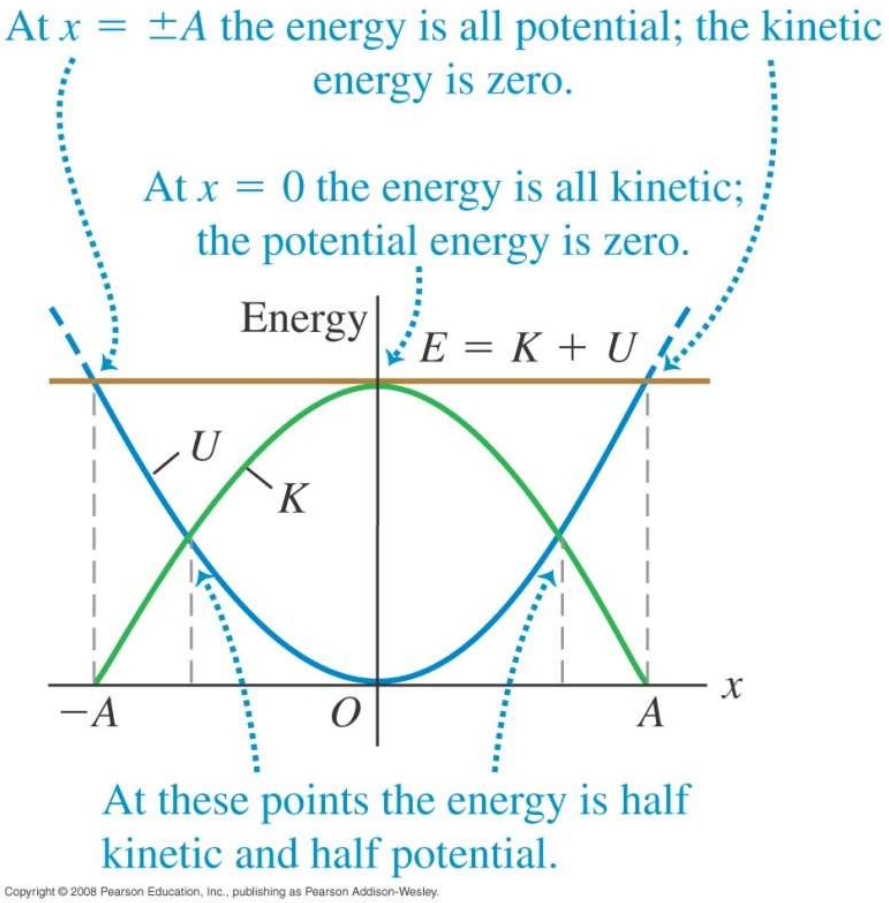
- At x = A (amplitude), the velocity is zero.



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Energy in SHM of a spring-mass system

- The below figure shows the potential energy (U) and kinetic energy (K) as function of position x for SHM.



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Example 4

Referring to example 2 where k = 200 N/m, m = 0.50 kg and the oscillating mass is released from rest at x = 0.020 m.

a) Find the maximum and minimum speed attained by oscillating body.

b) Find the maximum acceleration.

c) Determine the velocity and acceleration when the body has moved halfway to the centre from its original position.

d) Find the potential, kinetic and total energy at this position.

$$v_x = \pm\sqrt{\frac{k}{m}}\sqrt{A^2 - x^2}$$

$$v_{max}(at\,x = 0) = \sqrt{\frac{k}{m}}A = 0.40\ m/s$$

$$v_{min}(at\,x = A) = 0\ m/s$$

Example 4 - cont

$$a_{max}(at\ x = -A) = -\frac{k}{m}(-A)$$

$$= 8.0\ m/s^2$$

$$\text{At } x = \frac{A}{2} = 0.010\ m$$

$$v_x = -0.35\ m/s$$

$$a_x = -4.0\ m/s^2$$

$$\text{Total } E = \frac{1}{2}kA^2 = 0.040\,J$$

$$PE = \frac{1}{2}kx^2 = 0.010\,J$$

$$KE = \frac{1}{2}mv_x^2 = 0.030\,J$$

The simple pendulum – another example of SHM

- A simple pendulum consists of a point mass suspended by a massless un-stretchable string.
- If the point mass is displaced from equilibrium position, oscillations take place.
- The restoring force is provided by the component of the weight of the mass.
- The path of the point mass (also known as the pendulum bob) is an arc of circle with radius L which is the length of the string.

See animation

The simple pendulum

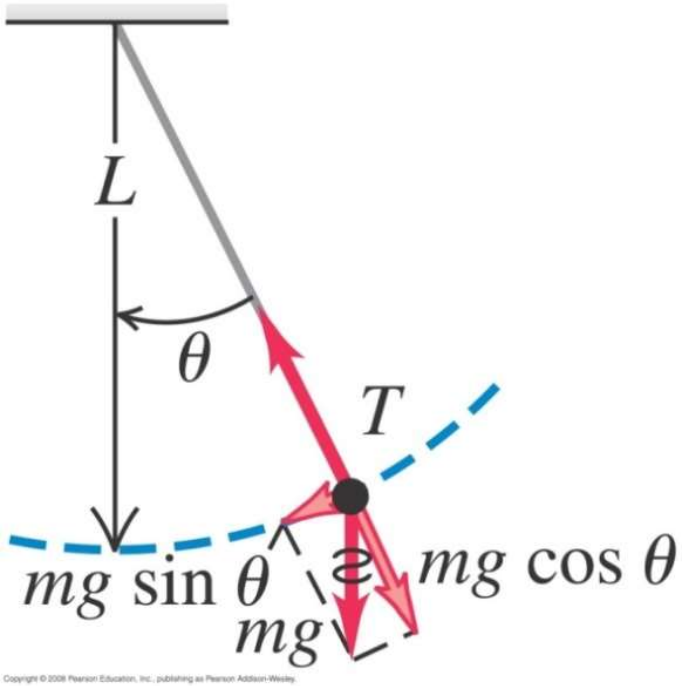
- From the figure $F = -mg\sin\theta$

- If θ is small, then $\theta \approx \sin\theta$.

$$F = ma = -mg\sin(\theta)$$

$$F = -mg\sin(\theta) = -mg(\theta) = -mg\frac{x}{L}$$

$$a = -g\frac{x}{L}$$



- This is similar to F = –kx where k is mg/L in this case. The angular frequency is therefore

$$\omega = \sqrt{\frac{mg/L}{m}} = \sqrt{\frac{g}{L}}$$

The simple pendulum

- The frequency and time period of a simple pendulum are

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$$

$$T = \frac{1}{f} = 2\pi\sqrt{\frac{L}{g}}$$

- The frequency and period depend only on the length of the string and acceleration due to gravity.

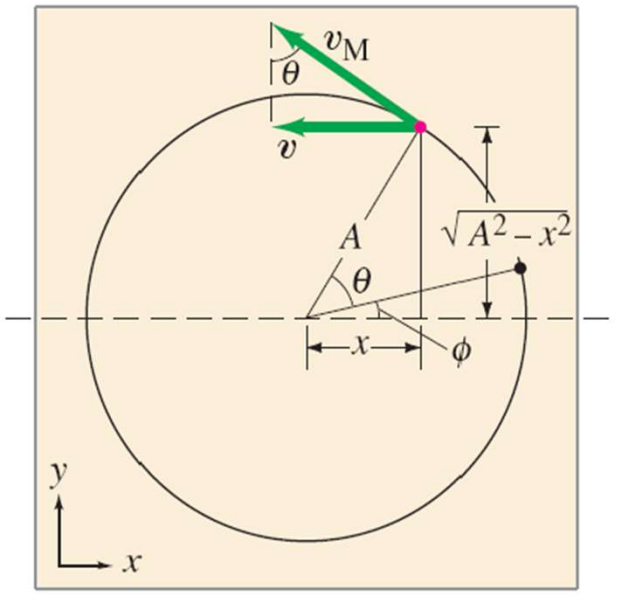
- Note that the above analysis is applicable only when the displacement and hence the amplitude is small.

Example 5

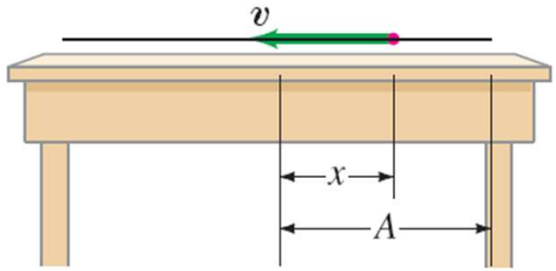
Find the period and frequency of a simple pendulum 1.000 m long at a location where acceleration due to gravity is 9.800 m/s^2.

Simple Harmonic Motion Related to Uniform Circular Motion

If we look at the projection onto the x or y axis of an object moving in a circle at a constant speed, we find that the x or y component of its position is the same as SHM.







Damped oscillations

- In the real world, dissipative forces cause a decrease in the amplitude of an oscillatory system.

- The simplest case is F = -bv, where b describes the strength of damping which occur in friction involving viscous fluid flow.

- The net force on the body is

$$F = ma = -kx - bv$$

$$m\frac{d^2x}{dt^2} = -kx - b\frac{dx}{dt}$$

Damping animation

Small damping

- Solving the differential equation, we get expressions is for an oscillator with small damping also known as **under damping**.

$$x = Ae^{(-b/2m)t}\cos(\omega' t + \phi)$$

$$\text{where } \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

- The angular frequency is smaller than in no damping case.
- The amplitude is a function of time and is not a constant.
- The amplitude of the oscillation decreases according to

$$A = A_o e^{(-b/2m)t}$$

Critical damping

- When the damping is increased further, a point is reached when ω' becomes zero, i.e.

$$\omega' = \frac{k}{m} - \frac{b^2}{4m^2} = 0$$

$$b = 2\sqrt{mk}$$

- The system is said to be **critically damped** and no longer oscillates.

- It returns to its equilibrium position when it is displaced and released.

Damping animation

Overdamping

- If $b > 2\sqrt{mk}$, the system is **over damped**.

- There is no oscillation and the system returns to equilibrium more slowly than in the critical damping case.

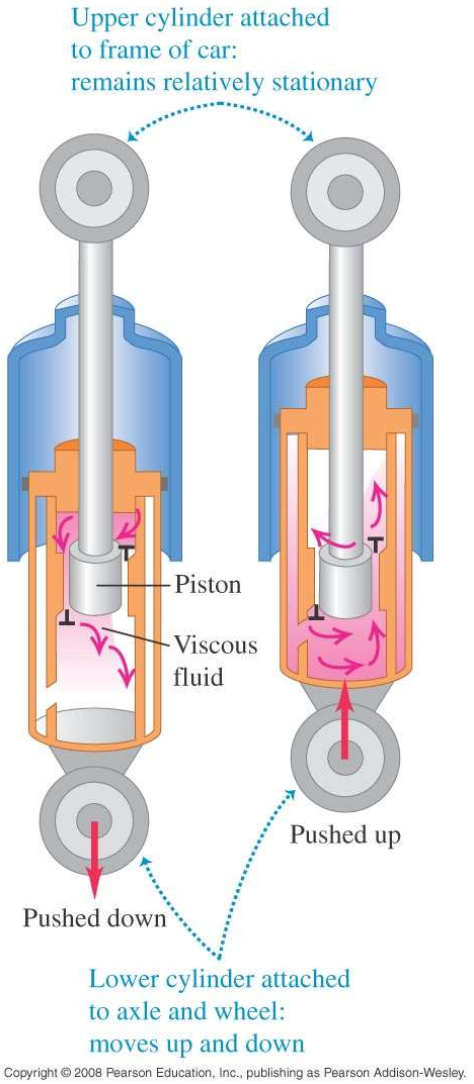
- The solution of the differential equation is

$$x = C_1 e^{-a_1 t} + C_2 e^{-a_2 t}$$

where C_1 and C_2 depend on initial conditions while a_1 and a_2 are determined by m, k and b.

Uses of damping

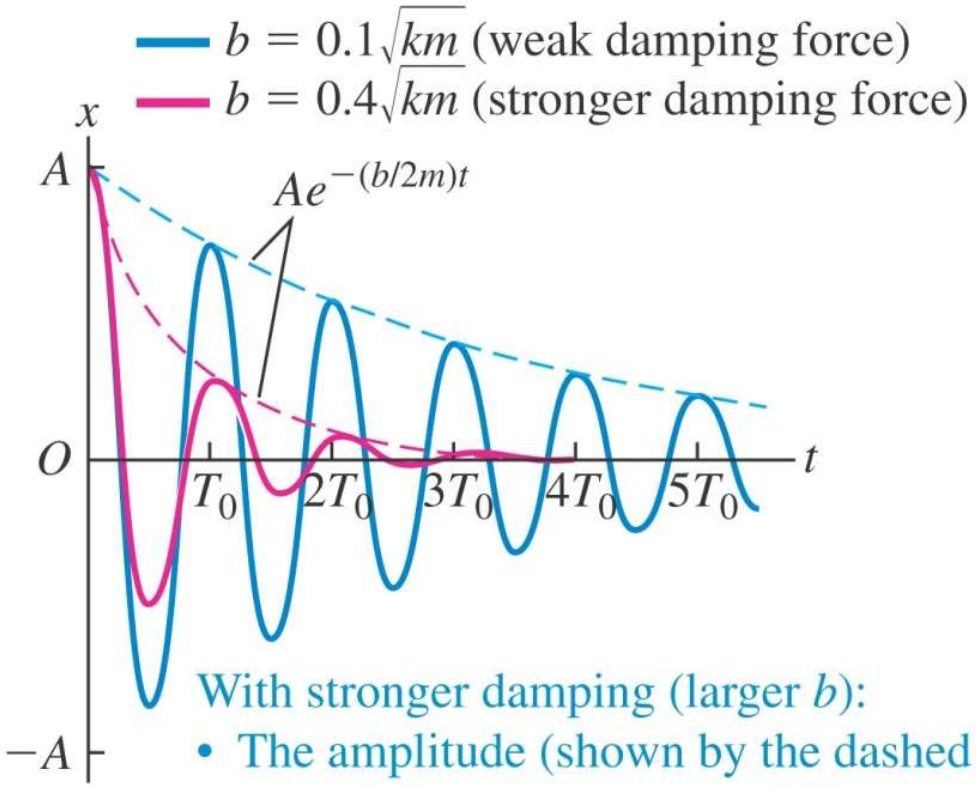
- One use of damping is in an automobile's suspension system.
- The shock absorbers provide a velocity dependent damping force so that when a car goes over a bump, it does not continue bouncing forever.
- For optimal passenger comfort, the system should be critically damped or slightly under damped.



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Graphs of displacement in damped oscillation

- For optimal passenger comfort, the system should be critically damped or slightly underdamped.



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Energy in damped oscillations

- For a spring-mass system

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

Differentiating w.r.t. time gives

$$\frac{dE}{dt} = mv\frac{dv}{dt} + kx\frac{dx}{dt}$$

$$= mva + kxv$$

$$= v(ma + kx) = v(-bv) = -bv^2$$

(from damped oscillations, $ma = -kx - bv$)

- The **rate of change of energy is negative** which means the energy decreases as the body oscillates.

Forced oscillations

- Periodically applying a driving force with angular frequency (ω_d) to a damped harmonic oscillator results in a **forced or driven oscillation**.

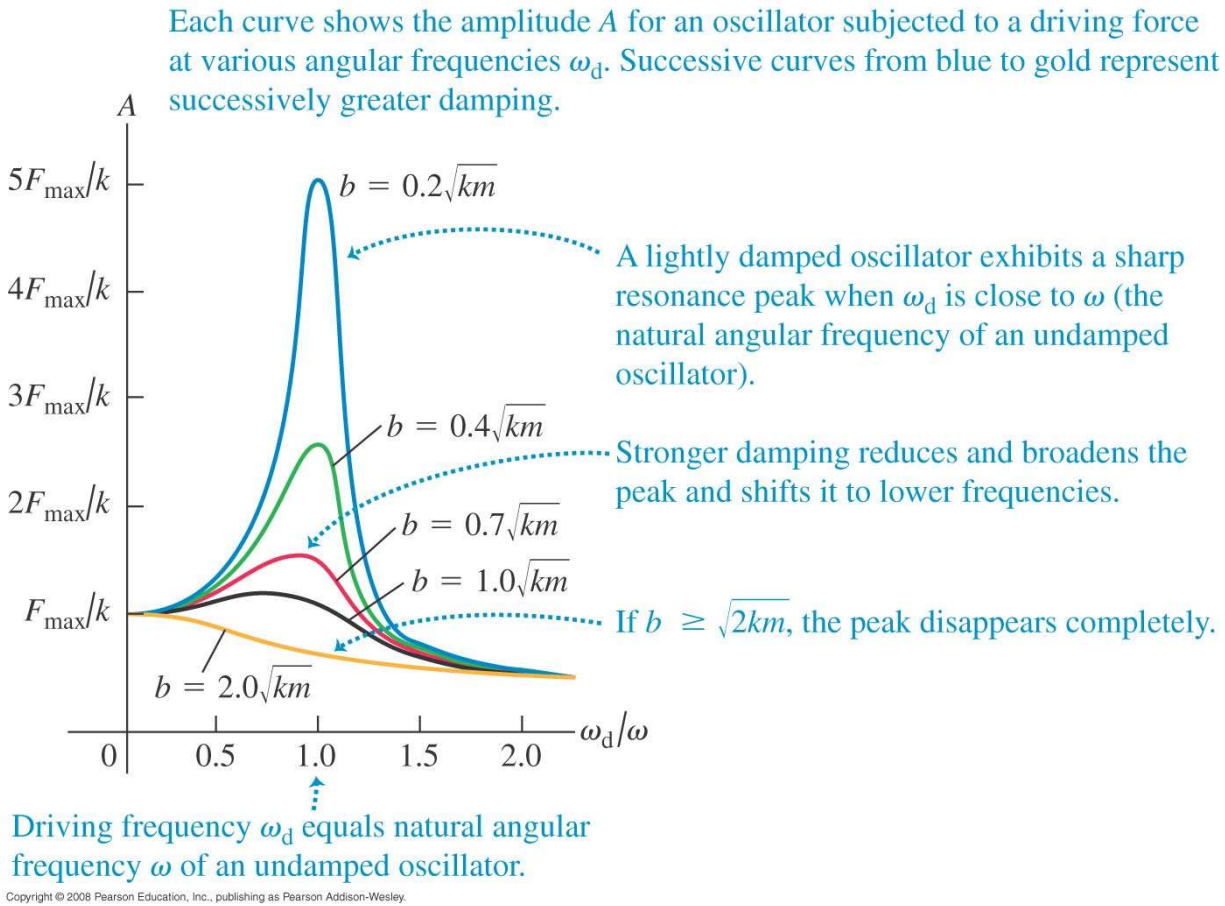
- The mass oscillates with angular frequency ω_d which may or may not be equal to the system's natural angular frequency ω'.

- The oscillation amplitude is a function of the driving frequency ω_d.

- When ω_d is equal to ω' the amplitude is highest and this is called **resonance.**

Amplitude at resonance

- An example of resonance is the building up of oscillations of a child on a swing by pushing it with a frequency equal to the swing's natural frequency.



Resonance of wine glass

End of chapter